

Dr. Doc:

1.Regarding the clinical manifestation of rhinitis, which of the following is TRUE?

- A. Nasal discharge is thick and purulent from the very onset of the illness.
- B. It is characterized by variable fever, inflammatory edema of the mucosa, and decreased secretions.
- C. Nasal discharge typically progresses from clear and watery to thick and purulent over 5 to 10 days.
- D. It is the least common manifestation of the common cold.

Answer: C

Explanation: In rhinitis, nasal discharge may be clear and watery at the onset, becoming thick and sometimes purulent as the infection progresses over a period of 5 to 10 days. It is also noted as the most common

2.All of the following are considered primary or common bacterial causes of pharyngitis EXCEPT:

- A. Streptococcus pyogenes
- B. Corynebacterium diphtheriae
- C. Neisseria gonorrhoeae
- D. Francisella tularensis

Answer: D

Explanation: While *S. pyogenes*, *C. diphtheriae*, and *N. gonorrhoeae* are listed as primary or major bacterial causes, *Francisella tularensis* is categorized as one of the other (rare) bacteria that have been implicated in acute pharyngitis.

3.Which statement regarding *Neisseria gonorrhoeae* as a pharyngeal pathogen is CORRECT?

- A. It is the most frequent bacterial cause of acute pharyngitis.
- B. It is typically transmitted via respiratory droplets during coughing.
- C. It is usually transmitted by oral–genital contact.
- D. It causes inflammation that is limited strictly to the nasal mucosa.

Answer: C

Explanation: *Neisseria gonorrhoeae* is a pharyngeal pathogen that is usually transmitted by oral–genital contact. *S. pyogenes* is actually the most frequent bacterial cause.

4. Which of the following is FALSE regarding the clinical presentation of pharyngitis and tonsillitis?

- A. They are characterized by pharyngeal pain, commonly known as a sore throat.
- B. The clinical appearance typically includes erythema (redness).
- C. Tissues usually appear pale and shrunken rather than swollen.
- D. Swelling of the affected tissues is a common clinical feature.

Answer: C

Explanation: pharyngitis and tonsillitis are characterized by pharyngeal pain and the clinical appearance of erythema and swelling of the affected tissues, not paleness or shrinkage.

5. In the context of URTIs, "mixed infections" refer to which of the following?

- A. An infection involving both the upper and lower respiratory tracts simultaneously.
- B. Infections that simultaneously involve two or more sites, such as rhinopharyngitis.

- C. An infection caused by a virus and a fungus at the same time.
- D. An infection that involves the lungs and the heart valves.

Answer: B

Explanation: Mixed infections are those that may simultaneously involve two or more sites, providing examples like rhinopharyngitis or tonsillopharyngitis.

6. Which of the following bacteria is identified as the most frequent bacterial cause of pharyngitis?

- A. *Corynebacterium ulcerans*
- B. *Arcanobacterium haemolyticum*
- C. *Streptococcus pyogenes*
- D. *Neisseria gonorrhoeae*

Answer: C

Explanation: The most (more than 80%) URTIs are viral, but among bacteria, *S. pyogenes* is the most frequent bacterial cause.

7. Which clinical feature is associated with rhinitis but NOT specifically mentioned as a hallmark of pharyngitis?

- A. Pharyngeal pain
- B. Erythema of tissues
- C. Nasal obstruction
- D. Tissue swelling

Answer: C

Explanation: Nasal obstruction is specifically listed as a clinical feature of rhinitis. While erythema and swelling occur in both, pharyngitis is primarily identified by sore throat/pharyngeal pain, whereas rhinitis is identified by nasal edema, secretions, and obstruction.

8.Regarding the biological properties of Rhinoviruses, which of the following is NOT CORRECT?

- A. They possess a linear, positive-sense ssRNA genome.
- B. They are acid-labile, which distinguishes them from many enteroviruses.
- C. They replicate more efficiently at 37°C than at 33°C.
- D. Their host range is extremely limited, infecting only humans and chimpanzees.

Answer: C

Explanation: The rhinoviruses grow better at 33°C, which is the temperature of the human nasopharynx. Replication at the higher core body temperature of 37°C is less efficient, which helps limit their primary pathology to the upper respiratory tract.

9.Which of the following statements about Adenovirus structure and classification is TRUE?

- A. Adenoviruses are enveloped viruses with a circular dsDNA genome.
- B. The icosahedral capsid is composed of exactly 252 capsomeres.
- C. The genus Aviadenovirus is responsible for most human respiratory infections.
- D. They are highly sensitive to bile salts and proteolytic enzymes in the GI tract.

Answer: B

Explanation: Adenoviruses are non-enveloped with a linear dsDNA genome and a capsid made of 252 capsomeres. Mastadenovirus infects mammals (including humans), while Aviadenovirus infects birds. They are notably resistant to bile salts and proteolytic enzymes, allowing them to replicate in the GI tract.

10. Which glycoprotein of the Coronavirus is the BEST ANSWER for mediating receptor binding and cell fusion?

- A. M (Membrane) protein
- B. HE (Hemagglutinin-acetylcholinesterase) protein
- C. N (Nucleocapsid) protein
- D. S (Spike) protein

Answer: D

Explanation: The S (Spike) protein is the large, club-shaped peplomer that acts as the major antigen and is specifically responsible for receptor binding and cell fusion.

11. All of the following are characteristics of Coronaviruses EXCEPT:

- A. They possess the largest genome among all RNA viruses.
- B. Their genome is segmented, allowing for frequent antigenic shift.
- C. They cause a slow, patchy destruction of ciliated epithelial cells.
- D. They are highly species-specific in their tropism.

Answer: B

Explanation: Coronaviruses have a non-segmented, positive-sense ssRNA genome. Because they are non-segmented, they do not undergo antigenic shift via reassortment like the Influenza virus.

12.Regarding Parainfluenzaviruses, which statement is CORRECT?

- A. They have a segmented genome similar to Influenza viruses.
- B. Type 4 is the most common cause of severe pneumonia in infants.
- C. Types 1 and 2 are primary causes of "croup" (laryngotracheobronchitis).
- D. They are antigenically unstable due to frequent point mutations.

Answer: C

Explanation: Parainfluenzaviruses are enveloped and non-segmented, making them antigenically stable. Types 1 and 2 are specifically associated with croup, causing respiratory obstruction through laryngeal swelling.

13.Which of the following statements about Influenza C viruses is FALSE?

- A. They contain 7 segments of RNA.
- B. They lack a neuraminidase gene.
- C. They are a common cause of global pandemics through antigenic shift.
- D. They can infect both humans and swine.

Answer: C

Explanation: Influenza B and C viruses do not exhibit antigenic shift. Additionally, Influenza C contains 7 RNA segments and lacks the neuraminidase gene.

14. In the context of Influenza virus nomenclature, what does the "15" represent in A/Swine/Iowa/15/30(H1N1)?

- A. The year of isolation.
- B. The HA subtype number.
- C. The strain number.
- D. The geographic origin code.

Answer: C

Explanation: The nomenclature system follows: Type / Host / Location / Strain Number / Year (Subtype). Therefore, "15" is the strain number and "30" (1930) is the year of isolation.

15. Which of the following is the BEST description of "Antigenic Drift" in Influenza?

- A. Sudden changes in viral surface antigens due to genetic reassortment.
- B. Minor antigenic changes caused by the accumulation of point mutations.
- C. The process by which the virus acquires a new lipid envelope.
- D. Co-infection of a cell by two different virus types.

Answer: B

Explanation: Antigenic drift involves minor changes resulting from point mutations in the genes for HA or NA, allowing the virus to slowly escape host immunity.

16. Which statement regarding the pathogenesis of Respiratory Syncytial Virus (RSV) is TRUE?

- A. Viremia is a common feature used for routine clinical diagnosis.
- B. It is a non-enveloped virus that is highly resistant to detergents.

- C. Replication begins in the nasopharynx and can spread to the lower tract via secretions.
- D. It is highly unstable antigenically, similar to Influenza A.

Answer: C

Explanation: RSV replication starts in the epithelial cells of the nasopharynx and spreads to the lower tract, likely via secretions. Viremia has not been detected and the virus is antigenically stable.

17. In Influenza pathogenesis, the role of Neuraminidase (NA) is to:

- A. Bind to sialic acid receptors to initiate viral entry.
- B. Lower the viscosity of the mucus film to expose cellular receptors.
- C. Inhibit host cellular messenger RNA translation.
- D. Surround the viral core to promote particle morphogenesis.

Answer: B

Explanation: Neuraminidase cleaves sialic acid, which lowers the viscosity of the mucus film, laying bare receptors and promoting the spread of the virus to the lower respiratory tract.

18. Which of the following Influenza proteins is correctly matched with its function?

- A. M1 Protein: Forms a membrane channel and is the target for amantadine.
- B. NS1 Protein: Inhibits host cellular messenger RNA translation.
- C. NP Protein: Cleaves sialic acid to promote virus release.
- D. PB1/PB2/PA: Major components of the lipid bilayer membrane.

Answer: B

Explanation: NS1 is a non-structural protein that inhibits cellular mRNA translation. M1 is a structural matrix protein. M2 (not M1) is the target for amantadine. NP forms the nucleocapsid.

19. Antigenic Shift in Influenza A is primarily possible because:

- A. The virus uses a reverse transcriptase with a high error rate.
- B. The viral genome is segmented into 8 pieces of RNA.
- C. The virus can infect humans, birds, and swine simultaneously.
- D. The M1 protein is highly susceptible to point mutations.

Answer: B

Explanation: Antigenic shift (major changes/new subtypes) occurs because of the segmented nature of the genome; when a cell is coinfecting by two different viruses, segments can be mixed/reassorted in progeny virions.

20. Why do Influenza B and C NOT cause pandemics through antigenic shift?

- A. Their genomes are non-segmented.
- B. They lack the hemagglutinin spike.
- C. They have very few related viruses existing in animals for reassortment.
- D. They do not infect the respiratory tract of humans.

Answer: C

Explanation: Antigenic shift requires reassortment with other strains, often from animals. Influenza B and C do not exhibit shift because few related viruses exist in animals to provide the necessary genetic diversity for reassortment.

Which part of the respiratory system is typically considered sterile due to the action of the ciliary escalator?

- a) Pharynx
- b) Nasal cavity
- c) Lower respiratory system (larynx, trachea, bronchi, alveoli)
- d) Middle ear

✓ Answer: c

Explanation:

▼ While the upper respiratory tract is continuously exposed to pathogens and has normal microbiota, the lower respiratory system is usually sterile due to the action of the mucociliary clearance system.

How does the normal microbiota of the nasal cavity and throat primarily protect the host?

- a) By producing stomach acid
- b) By competing with pathogens for nutrients and attachment sites
- c) By secreting mucus
- d) By inducing high fever

✓ Answer: b

Explanation:

▼ Resident flora prevents colonization by pathogenic microbes through microbial antagonism and competition for nutrients and attachment sites.

Which hemolytic pattern is characterized by the complete lysis of red blood cells on blood agar?

- a) Alpha (α)
- b) Beta (β)
- c) Gamma (γ)
- d) Delta (δ)

✔ Answer: b

Explanation:

Beta-hemolysis produces a clear zone around colonies due to complete destruction of red blood cells.

Streptococcus pyogenes is classified under which Lancefield group?

- a) Group A
- b) Group B
- c) Group D
- d) Not applicable

Answer: A

Explanation:

S. pyogenes is the major human pathogen belonging to Lancefield Group A.

Which diagnostic feature is used to differentiate *Streptococcus pneumoniae* from other alpha-hemolytic streptococci?

- a) Bacitracin sensitivity
- b) Optochin sensitivity and bile solubility
- c) Growth in 6.5% NaCl
- d) Catalase positivity

Answer: B

Explanation:

▼ *S. pneumoniae* is optochin-sensitive and bile-soluble, whereas viridans streptococci are generally optochin-resistant and bile-insoluble.

A 2-year-old child is brought to the clinic due to recurrent acute otitis media. Which anatomical characteristic of the pediatric Eustachian tube primarily predisposes young children to middle ear infections compared to adults?

- A) Longer, narrower, and more vertical
- B) Shorter, wider, and more horizontal
- C) Completely cartilaginous along its entire length

- D) Lacks a mucosal lining

Correct Answer: B) Shorter, wider, and more horizontal

Explanation: In infants and young children, the Eustachian tube is shorter, wider, and oriented more horizontally (at an angle of $\sim 10^\circ$ compared to $\sim 45^\circ$ in adults). This anatomical layout prevents efficient passive drainage and allows nasopharyngeal secretions, viruses, and bacteria to easily reflux into the middle ear cavity.

An upper respiratory tract infection causes swelling of the nasopharyngeal mucosa, leading to complete obstruction of the Eustachian tube. What is the primary mechanism that initiates fluid accumulation (transudate) in the middle ear cavity?

- A) Active bacterial secretion of exotoxins into the middle ear
- B) Absorption of trapped air by mucosal capillaries, creating a negative pressure (vacuum)
- C) Direct rupture of the internal carotid artery
- D) Failure of the tensor tympani muscle to contract

Correct Answer: B) Absorption of trapped air by mucosal capillaries, creating a negative pressure (vacuum)

Explanation: When the Eustachian tube is blocked, air is trapped inside the middle ear. The mucosal lining continuously absorbs oxygen and nitrogen into the bloodstream, creating a negative pressure (vacuum) inside the sealed cavity. This negative pressure "sucks" serous fluid (transudate) out of mucosal blood capillaries into the middle ear space.

A 3-year-old girl is brought in for a routine check-up 4 weeks after successfully completing antibiotics for Acute Otitis Media (AOM). She has no fever, no ear pain, and is playing normally. Otoscopy shows amber-colored fluid behind an intact, non-bulging tympanic membrane with a visible fluid level. What is the most appropriate management?

- A) Immediate oral Amoxicillin for 10 days
- B) Emergent tympanocentesis
- C) Observation and re-evaluation in 3 months
- D) High-dose oral Prednisone

Correct Answer: C) Observation and re-evaluation in 3 months

Explanation: This patient has Otitis Media with Effusion (OME) (fluid in the middle ear without signs of acute inflammation like fever, bulging, or pain). Post-AOM effusions are very common and usually resolve spontaneously within 3 months. Antibiotics and steroids are not indicated for uncomplicated OME.

Which physical examination finding on pneumatic otoscopy is considered the most reliable and specific indicator of middle ear effusion in Acute Otitis Media?

- A) Erythema of the external ear canal
- B) Decreased / absent mobility of the tympanic membrane

- C) Presence of cerumen in the ear canal
- D) Tenderness when pulling the pinna

Correct Answer: B) Decreased / absent mobility of the tympanic membrane

Explanation: Pneumatic otoscopy assesses eardrum mobility. Trapped fluid/pus in the middle ear creates back-pressure that prevents the tympanic membrane from moving when air is puffed against it. Decreased or absent mobility is the single most sensitive and specific finding for middle ear effusion.

What is the single most common bacterial pathogen responsible for Acute Otitis Media in pediatric patients?

- A) *Pseudomonas aeruginosa*
- B) *Streptococcus pneumoniae*
- C) *Moraxella catarrhalis*
- D) *Staphylococcus aureus*

Correct Answer: B) *Streptococcus pneumoniae*

Explanation: The classic triad of bacterial pathogens in AOM (in order of frequency) is:

1. *Streptococcus pneumoniae* (~35–40%)

2. Non-typable *Haemophilus influenzae* (~30–35%)

3. *Moraxella catarrhalis* (~15–20%)

A 14-month-old boy presents with a 2-day history of fever, right ear pulling, and bilateral purulent conjunctivitis ("pink eye" with thick discharge). Which organism is classically associated with this "otitis-conjunctivitis syndrome"?

- A) *Streptococcus pyogenes* (Group A Strep)
- B) Non-typable *Haemophilus influenzae*
- C) *Mycoplasma pneumoniae*
- D) *Pseudomonas aeruginosa*

Correct Answer: B) Non-typable *Haemophilus influenzae*

Explanation: Non-typable *H. influenzae* is uniquely associated with Otitis-Conjunctivitis Syndrome, where a patient presents simultaneously with acute otitis media and purulent bacterial conjunctivitis.

A 18-month-old infant is diagnosed with uncomplicated Acute Otitis Media. He has no known drug allergies and has not received antibiotics in the past 30 days. What is the recommended first-line antibiotic treatment?

- A) High-dose oral Amoxicillin (80–90 mg/kg/day)

- B) Oral Ciprofloxacin
- C) Oral Azithromycin
- D) Intravenous Vancomycin

Correct Answer: A) High-dose oral Amoxicillin (80–90 mg/kg/day)

Explanation: High-dose Amoxicillin (80–90 mg/kg/day) is the first-line antibiotic of choice for AOM because it achieves high concentrations in the middle ear fluid to overcome penicillin-resistant *S. pneumoniae*.

A 2-year-old child was started on high-dose Amoxicillin 72 hours ago for AOM. Today, his mother returns because he still has a high fever (39.1°C), severe ear pain, and a persistently bulging tympanic membrane. What is the most appropriate next step in management?

- A) Continue Amoxicillin for another 7 days
- B) Switch to oral Amoxicillin-Clavulanate (Augmentin)
- C) Switch to oral Erythromycin
- D) Perform an immediate mastoidectomy

Correct Answer: B) Switch to oral Amoxicillin-Clavulanate (Augmentin)

Explanation: Treatment failure after 48–72 hours of Amoxicillin suggests infection by a β -lactamase-producing organism (such as non-typable *H. influenzae* or *M. catarrhalis*). Adding Clavulanate (a β -lactamase inhibitor) restores antimicrobial coverage.

During otoscopic examination of a inflamed, bulging eardrum in AOM, the normal "cone of light" (light reflex) is described as "lost" or "distorted." What is the underlying physical reason for this phenomenon?

- A) The ear canal is filled with thick cerumen
- B) The tympanic membrane surface has lost its smooth, reflective contour due to swelling and middle ear pressure
- C) The ossicles have detached from the tympanic membrane
- D) The light bulb of the otoscope is malfunctioning

Correct Answer: B) The tympanic membrane surface has lost its smooth, reflective contour due to swelling and middle ear pressure

Explanation: A healthy eardrum is smooth, translucent, and concave, reflecting otoscope light as a sharp, triangular "cone of light." Inflammation, cloudiness, and outward bulging alter the surface angle, causing the light to scatter randomly, resulting in a loss or distortion of the light reflex.

A 5-year-old child presents with a sore throat, fever, difficulty swallowing, and a highly swollen neck described as 'bull neck'. Upon visual examination of the pharynx, the physician observes a thick, greyish-white, tough membranous coating that covers the tonsils and pharynx. Attempting to scrape this membrane causes immediate bleeding. What is the most likely diagnosis and causative organism?

- A. Whooping cough; caused by *Bordetella pertussis*.
- B. Primary atypical pneumonia; caused by *Mycoplasma pneumoniae*.
- C. Acute epiglottitis; caused by *Haemophilus influenzae* type b.
- D. Streptococcal pharyngitis; caused by *Streptococcus pyogenes*.
- E. Pharyngeal diphtheria; caused by toxigenic *Corynebacterium diphtheriae*.

Answer:E